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TEXT EXTRACTİON FROM AN IMAGE

ADVANCED COMPUTER VİSİON AND IMAGE PROCESSİNG

Start

Introduction

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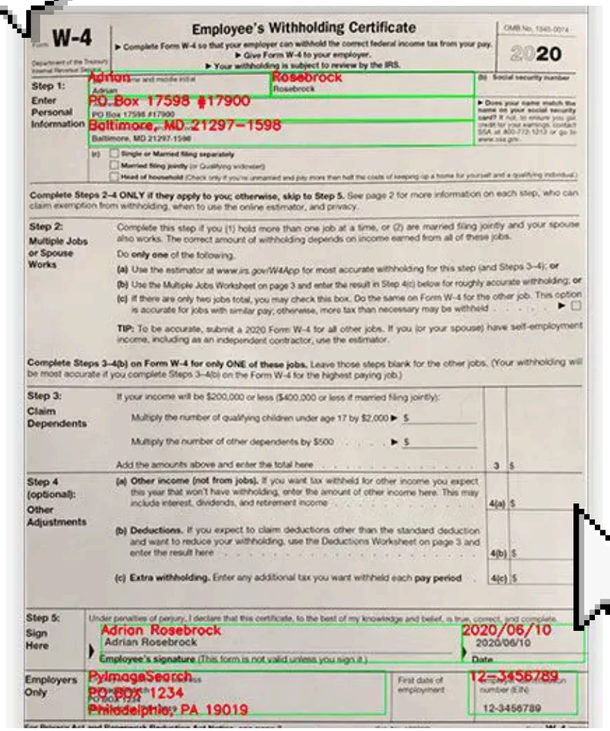

-Text extraction from images is a key task in document digitization and intelligent information processing. It enables converting visual text into machine-readable text for further analysis and storage.

-Example of converting textual content from an image into machine-readable text using OCR techniques.

Problem Statement

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-Extracting text from images is challenging due to low image quality, noise, different fonts, lighting conditions, and complex backgrounds.

-Example of image quality issues such as blur, noise, and poor illumination that negatively affect text extraction accuracy.

Objectives

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*Extract text accurately from images
*Handle noisy and low-quality images
*Combine detection and recognition techniques
*Produce readable and structured output

-Illustration of the main objectives of the proposed system, focusing on accurate and automated extraction of textual information from images.

Related Work

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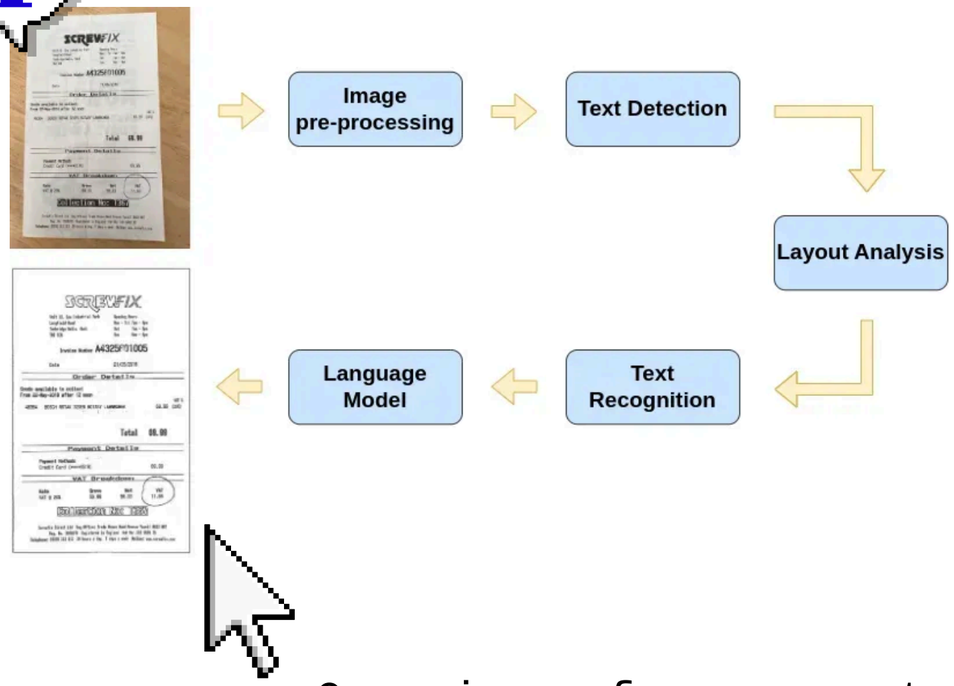

-Several deep learning approaches have been proposed for text detection and recognition, such as CRAFT,EAST, and Tesseract-based systems.

-Overview of representative approaches for scene text detection and recognition reported in previous studies.

Proposed Solution

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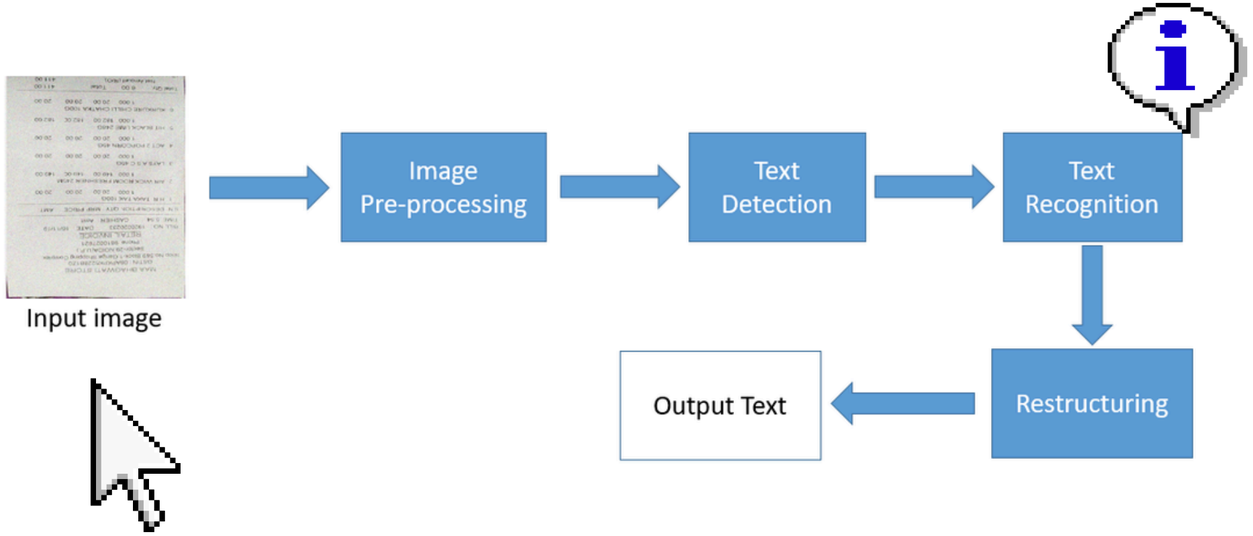
-Our system adopts a hybrid approach that combines deep learning-based text detection with OCR techniques to improve accuracy and robustness.



-Illustration of the main stages involved in the proposed text extraction pipeline.

System Architecture

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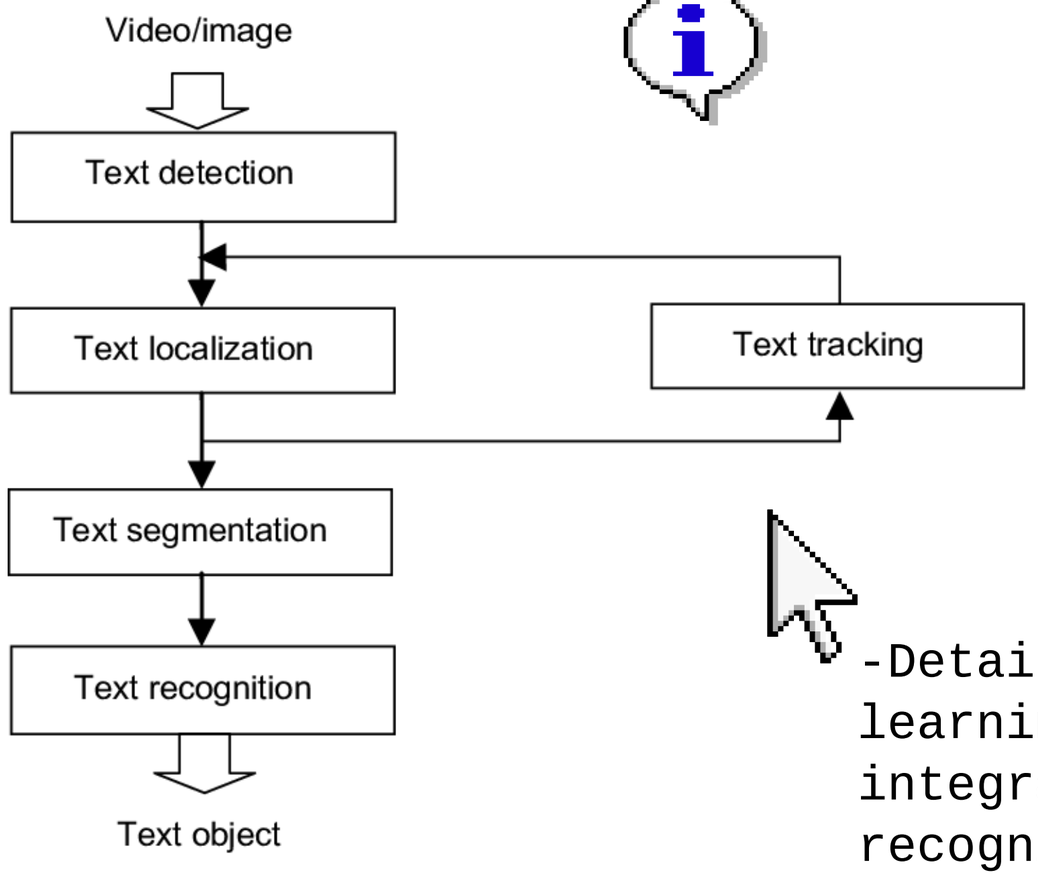

-The system processes the image through preprocessing, text detection, text recognition, and final text generation.

-Detailed architecture of a deep learning-based OCR system integrating text detection and recognition components.

Technologies Used

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*Python
*OpenCV

*Tesseract OCR
*Deep Learning Models (CRAFT, TrOCR)

System Implementation

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-The proposed system was implemented using a modular architecture that integrates image preprocessing, text detection, and text recognition components. The implementation ensures flexibility,scalability, and ease of use through a graphical user interface.

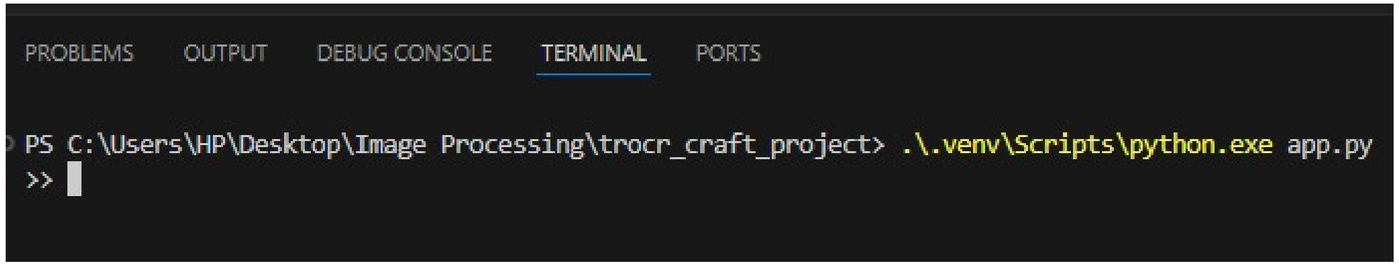

Figure (a): Source code implementation

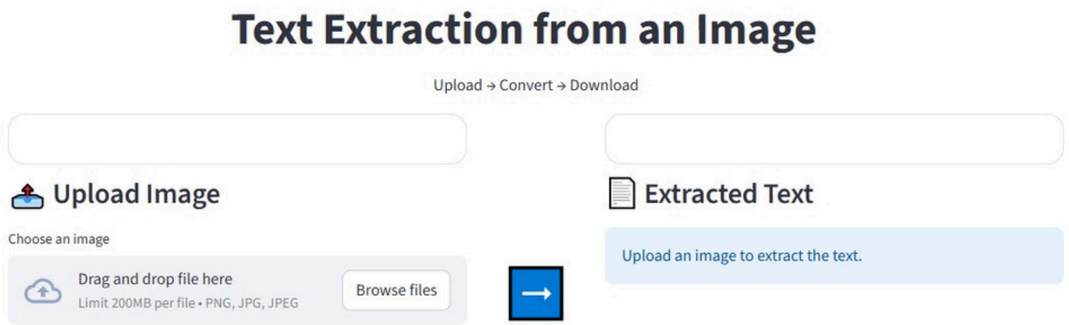

Figure (b): User interface before text extraction

Experimental Results

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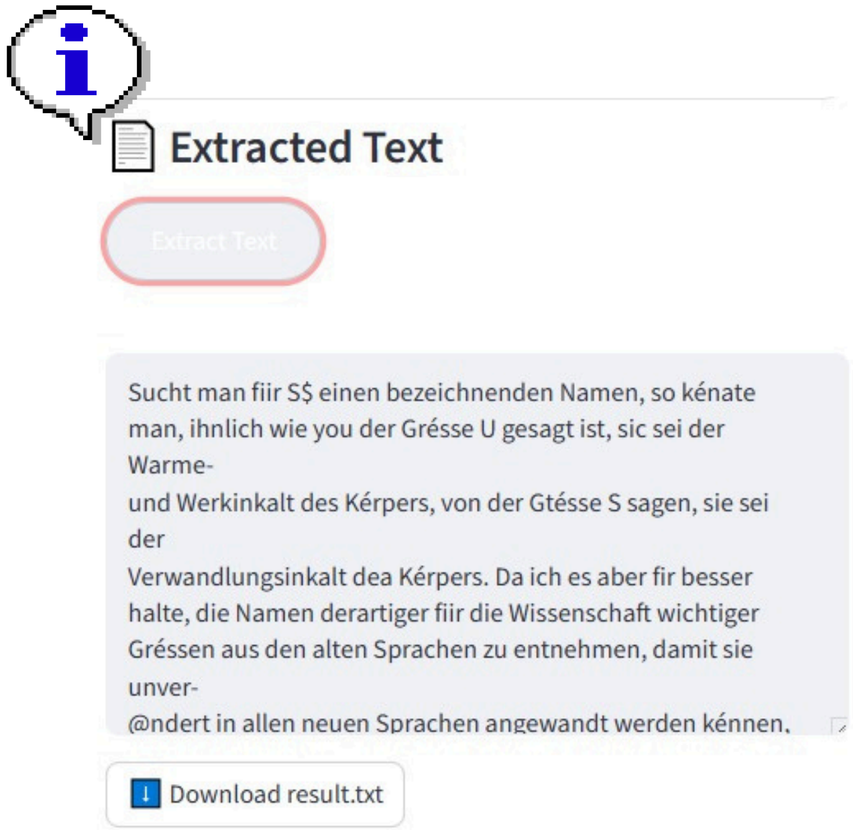

-This slide presents sample outputs generated by the proposed system. The results demonstrate the ability of the OCR pipeline to accurately extract textual content from different image types, including printed documents and receipts.

Performance Evaluation

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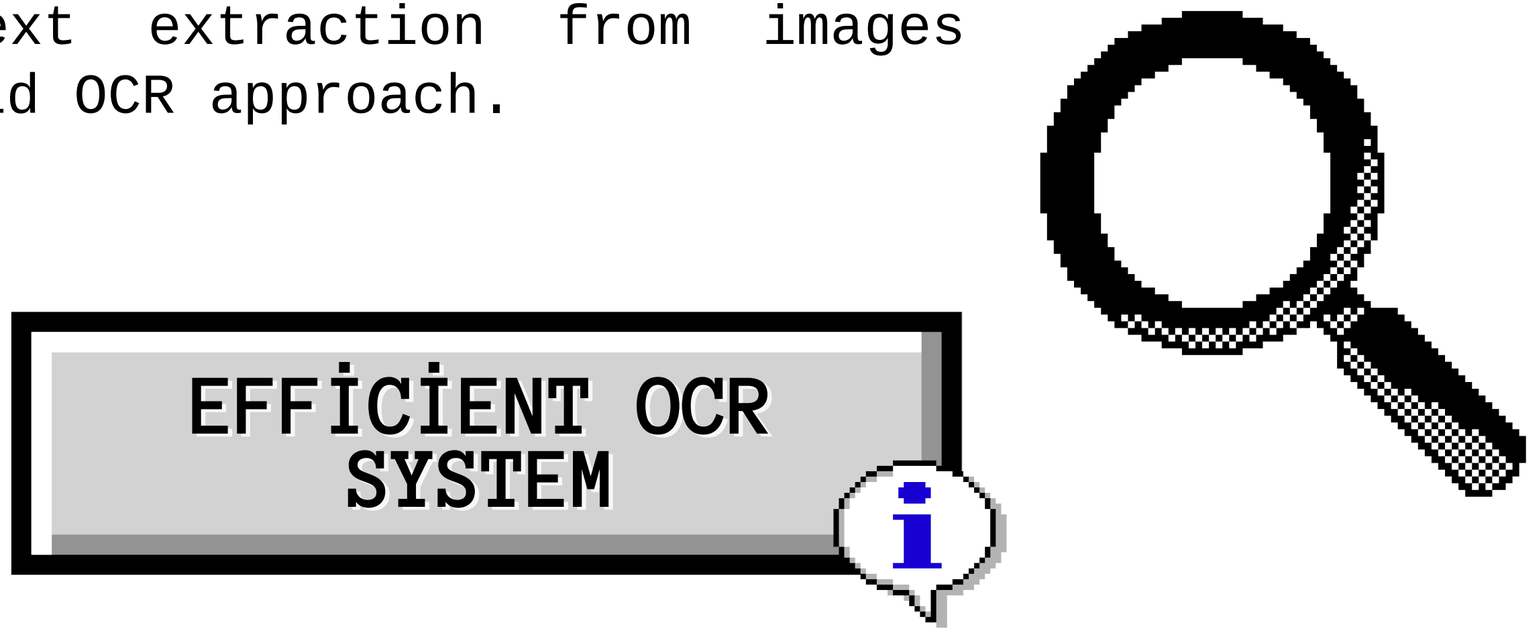
Metric	Value	Description
Accuracy	High (≈90%+)	Correctly recognized characters
Processing Time	~1–2 sec/image	End-to-end OCR pipeline
Model Size	Medium	Suitable for desktop deployment

-This slide summarizes the performance evaluation of the proposed OCR system. The system was assessed using accuracy, processing time, and model size as key performance indicators.

-The results show that the system achieves reliable text recognition accuracy while maintaining reasonable processing time, making it suitable for real-world document analysis tasks.

Conclusion

-The proposed system demonstrates effective text extraction from images using a hybrid OCR approach.



Future Work

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-Future improvements include multilingual OCR support, handwriting recognition, and mobile deployment.

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